ADRIÀ MITJAVILA VENTURA

MSc | PhD student in Bioinformatics at *Institut Josep Carreras*

Bioinformatician with over 5 years of experience in biological data analysis, statistical modelling and computational workflows. Background in Molecular Genetics and Biotechnology, with previous hands-on experience in microbiological studies, combining quantitative data analysis, experimental research and scientific reporting. Experienced in omics data analysis and integration, with a strong interest in applying my skills to the biotechnology industry. See more about me on my website.



EXPERIENCE

11/2022 | Present

PhD student

Institut Josep Carreras & *Institut Germans Trias i Pujol* — Badalona, Spain

Group of Regulatory Genomics (Dr. Tanya Vavouri)
Study of Piwi-interacting RNAs (piRNAs), their biogenesis mechanisms and their conservation across species. | Analysis and integration of bulk small RNA-seq, RNA-seq, ChIP-seq and other genomics data. | Analysis of single-cell RNA-seq and 10x Xenium spatial transcriptomics data. | Mentoring and outreach activities adressed to high-school and undergraduate students.

- Bioinformatics and statistical analysis (R, Bash, Python).
- Pipeline development (Snakemake).
- Workflow and environment management (conda, docker, singularity).
- Scientific writing.
- Version control (git, GitHub).
- High performance computing (Slurm).
- Outreach and mentoring.

08/2021 | 11/2022

Bioinformatician & master's thesis student

Institut Josep Carreras — Badalona, Spain

Group of Regulatory Genomics (Dr. Tanya Vavouri)
Study of Piwi-interacting RNAs (piRNAs), their biogenesis mechanisms and their conservation across species.

- Bioinformatics and statistical analysis (R, Bash, Python).
- Pipeline development (Snakemake).
- Workflow and environment management (conda, docker, singularity).
- Scientific writing.
- Version control (git, GitHub).
- High performance computing (Slurm).
- Outreach and mentoring.

07/2019 | 07/2021

Bioinformatician

Istituto Europeo di Oncologia — Milan, Italy

Group of Epigenetic Mechanisms in Cancer (Dr. Diego Pasini)
Study of Wnt signaling pathway -and others- and its effects in transcription and epigenetic modifications in the context of colorectal cancer and stem cell development. | Analysis and integration of distinct omics data (RNA-/ChIP-/ATAC-seq) | Comparison of different ATAC-seq peak callers | Comparison of different sequencing depths for RNA-seq for DE analysis.

- Bioinformatics and statistical analysis (R, Bash, Python).
- Pipeline development (Snakemake).
- Workflow and environment management (conda, docker, singularity).
- Version control (git, GitHub).
- High performance computing (PBS).

CONTACT

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/amitjavilaventura
0000-0002-0337-1929

SKILLS

Informatics:

R/Bioconductor, R Markdown, Shiny, Bash, Snakemake, Python, Git/GitHub, HPC (PBS & Slurm), Conda, Docker, Singularity, Linux (Ubuntu)

Languages:

Catalan (Native), Spanish (Bilingual), English (IELTS 7.5 - C1), Italian (Basic)

DEVELOPMENT

`plotmics`: visualization of omics and sequencing data in R.

`ggmitji`: complements and customization for ggplots.

`chromHMMviewR`: visualization of ChromHMM outputs in R.

More on my GitHub profile.

References available upon request.
Build with pagedown, code here.
Last updated on 2026-10-16.

09/2017 | 10/2018

Master's thesis student & Research assistant

Instituto de Biomedicina de Sevilla — Seville, Spain

Group of Gene Expression (Dr. Sebastián Chávez)
Study of Xrn1 and Ccr4 roles in yeast nucleosome architecture and gene transcription. | Analysis of MNase-seq data and MNase-qPCR experiments from Xrn1-, Ccr4-mutant and WT yeast strains.

- Bioinformatics and statistical analysis (Bash, R, Past).
- UCSC Genome Browser.
- DNA extraction.
- MNase-qPCR.
- Microbiological cultures.
- Scientific writing.

07/2017 | 09/2017

Internship student

Institut d'Investigació Biomèdica de Girona — Girona, Spain

Unit of Diabetes, Endocrinology and Nutrition
Extracurricular internship. Identification of novel factors (genetic and bacterial) related to adipose tissue physiology and obesity. | RNA/DNA extraction from mouse hepatic biopsies and human fecal material.

- DNA/RNA extraction & qPCR.

09/2016 | 06/2017

Internship & bachelor's thesis student

Microbial. Sistemes i aplicacions analítiques, S.L. — Girona, Spain

Microbiological and molecular analyses of food and water for pathogen detection and food forensics | Study of butyrate effects on the ileal microbiota of colectomized colorectal cancer patients.

- DNA extraction, qPCR & molecular cloning.
- Bacterial cultures.
- Statistical analysis of NGS data (SPSS).
- Scientific writing.
- Sample traceability and results documentation

EDUCATION

2022 | Present

PhD in Bioinformatics

Universitat Autònoma de Barcelona (UAB) — Barcelona, Spain

- INPhINIT Incoming PhD Fellowship from LaCaixa.
- Research carried out at Regulatory Genomics group of the Institut Josep Carreras (IJC).
- Supervisor: Dr. Tanya Vavouri.

2018 | 2022

MSc in Bioinformatics and Biostatistics

Universitat Oberta de Catalunya (UOC) — Online

- Focused on Computational Genomics.
- Master's thesis: *Analysis of the variation in piRNA expression across three close related Mus species.* Grade: 9.7/10 (*with honours*). Supervisor: Dr. Tanya Vavouri.
- Overall grade: 8.94/10.

2017 | 2018

MSc in Molecular Genetics and Biotechnology

Universidad de Sevilla (US) — Seville, Spain

- Specialization in Molecular Biology Research.
- Master's thesis: *Influence of Xrn1 in nucleosome positioning across Saccharomyces cerevisiae genome and its effects in transcription.* Grade: 9.2/10. Supervisors: Dr. Sebastián Chávez de Diego & Dr. Gabriel Gutiérrez Pozo.
- Overall grade: 9.33/10.

2012 | 2017

BSc in Biotechnology

Universitat de Girona (UdG) — Girona, Spain

- Specialization in Cellular & Molecular Biotechnology.
- Bachelor's thesis: *Butyrate-induced changes in the diversity of intestinal mucosa-associated microbiota in colorectal cancer patients submitted to a lateral ileostomy.* Grade: 9.4/10. Supervisor: Prof. Jesús García-Gil.
- Overall grade: 8.12/10.

PUBLICATIONS

2026 — Mechanisms seeding the emergence of new piRNAs.
Mitjavila-Ventura & Vavouri. *Trends in Genetics*. 2026. doi: 10.1016/j.tig.2026.04.004

2025 — Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse
Casas, Mitjavila-Ventura *et al.*, *EMBO Journal*, 2025. doi: 10.1038/s44318-025-00475-4

2021 — Xrn1 influence on gene transcription results from the combination of general effects on elongating RNA pol II and gene-specific chromatin configuration
Begley *et al.*, *RNA Biology*, 2021. doi: 10.1080/15476286.2020.1845504

POSTERS & PRESENTATIONS

12/2024 — Genetic polymorphisms lead to major, locus-specific, variation in piRNA production in mouse
Mitjavila-Ventura, Casas et al. Poster at the *Jornades de Bioinformàtica i Genòmica* from the *Societat Catalana de Biologia*.
Lleida, Spain

05/2023 — Fast evolution of piRNA cluster expression in mouse.
Mitjavila & Vavouri. Poster at the Microsymposium in RNA Biology.
Vienna, Austria

HONORS & AWARDS

2022 | 2025 — INPhINIT Incoming PhD Fellowship
La Caixa Foundation

2022 — Master's thesis, *with honours*
Universitat Oberta de Catalunya

2019 | 2021 — SEMM PhD Fellowship
Scuola Europea di Medicina Molecolare (declined)

2015 | 2021 — Elite athlete
Boletin Oficial del Estado

2016 — World Rowing U23 Champion
FISA - World Rowing

ORGANIZATIONS

2018 | Present — *Societat Catalana de Biologia* (SCB)
Member (2018-Present)

2017 | 2025 — *Associació de Biotecnòlegs de Catalunya* (ASBTEC)
Member (2017-2025), collaborator (2018-2025) and board member (2019-2023)